L2.5 Curve Sketching

Recall: L2.4 Limits at Infinity; Horizontal Asymptotes

Name _____ **Solo – 9 minutes**

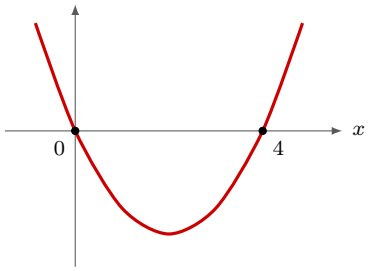
1. Evaluate $\lim_{x\to\infty} \dfrac{5x^2+3x-1}{2x^2-7}$.

2. Evaluate $\lim_{x\to-\infty} \dfrac{\sqrt{4x^2+1}}{3x-2}$.

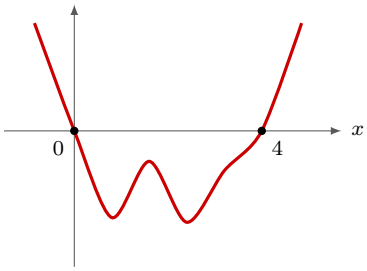
3. Everything below is true of the same function.

D all real numbers
I zeroes at $x=0$ and $x=4$, and nowhere else; y-intercept 0
A no vertical or horizontal asymptotes. **EB**: $f(x)\to+\infty$ at both ends
S neither even nor odd

Both sketches below match every test above. What would the derivatives table have to show for A to be the real graph? For B?



A



B